

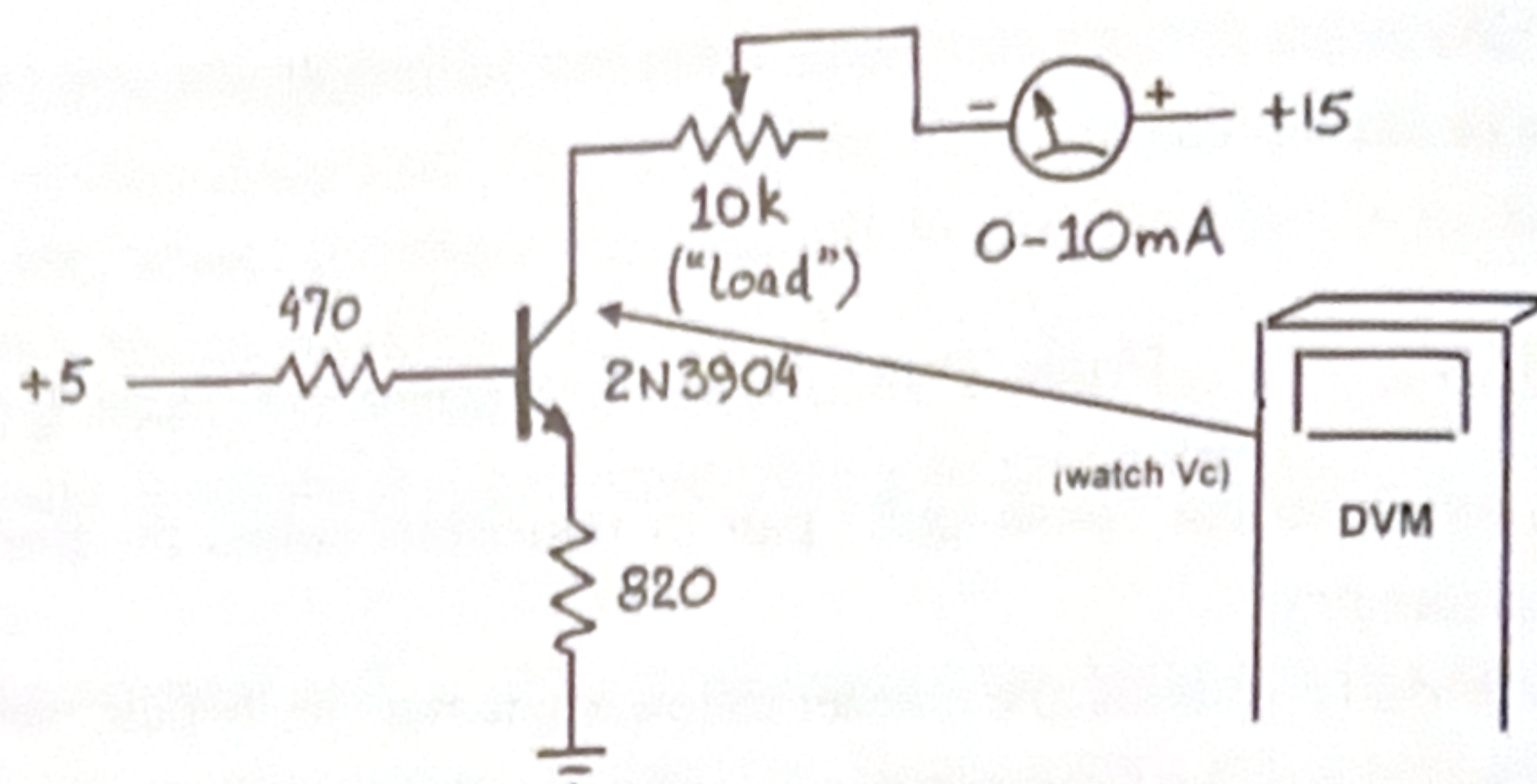


Figure 4L.5 Transistor current source.

happens at maximum resistance? Can you explain, in terms of *voltage compliance* of the current source (“compliance” is jargon for “range in which the circuit works properly”)?¹¹

Even within the compliance range, there are detectable variations in output current as the load is varied. What causes these variations?¹²

4L.5 Common-emitter amplifier

Wire up the common-emitter amplifier shown in Fig. 4L.6. What should its voltage gain be? Check it out. Is the signal’s phase inverted?

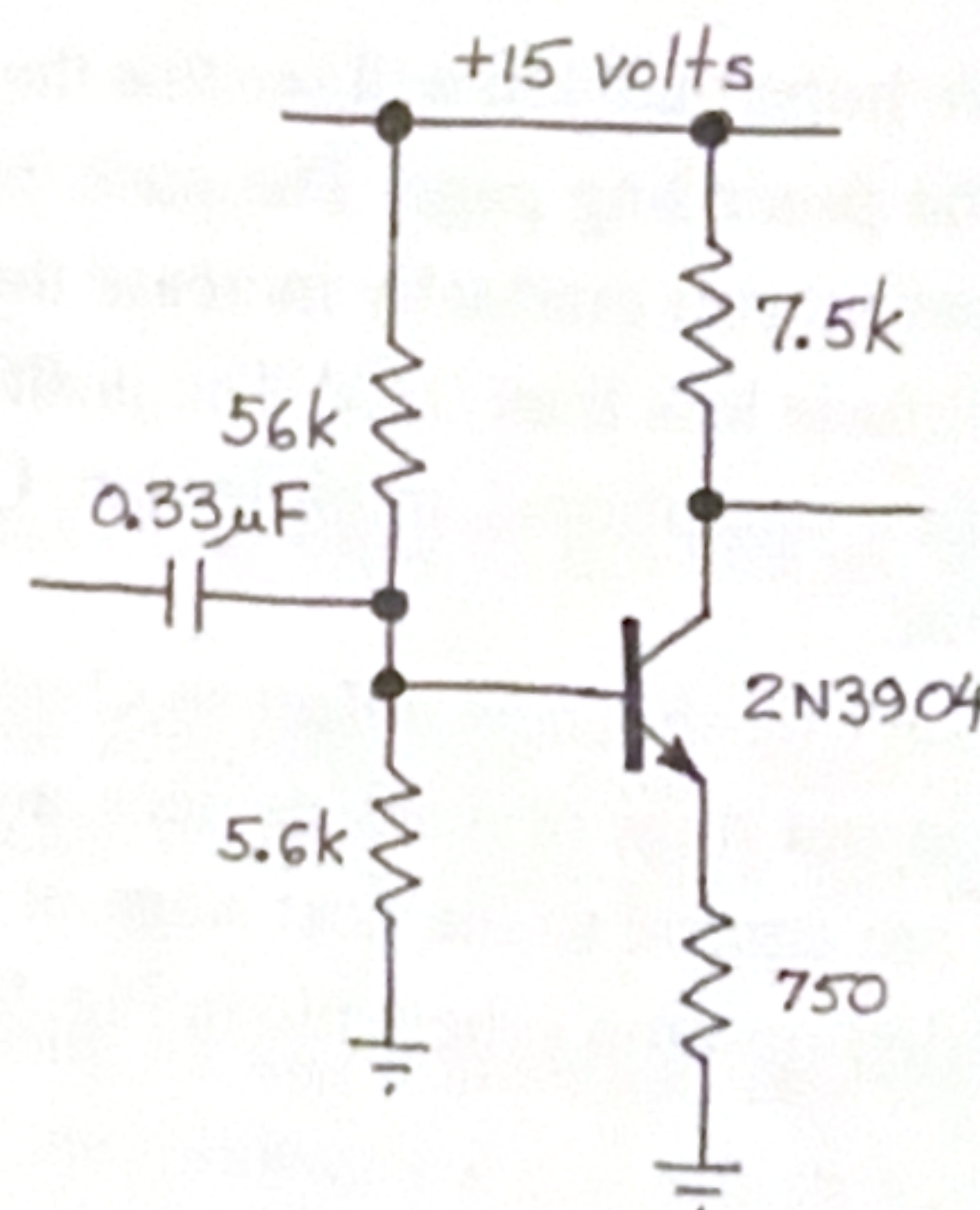


Figure 4L.6 Common-emitter amplifier.

Is the collector *quiescent* operating point correct (that is, its resting voltage)?¹³ How about the amplifier’s low frequency 3 dB point?¹⁴ What should the output impedance be?¹⁵ Check it by connecting a resistive load – let’s say 7.5k – using a blocking capacitor. (The blocking cap, again, lets you test impedance at signal frequencies without messing up the biasing scheme.)

- ¹¹ The current source fails when the transistor saturates. The DVM will let you measure or infer V_{CE} . Thus you can look for the relation between saturation and the fall of I_{out} .
- ¹² Early effect is the principal cause: varying V_{CE} slightly varies the effective base width. See §5S.2.
- ¹³ What level would be “correct?” The voltage that permits the widest swing without “clipping” – hitting either upper or lower limit. Roughly, that quiescent point would be half the supply voltage. A perfectionist would say, “Better to put it midway between the lower limit of about 1 V (V_E) and the supply. But with a 15 V supply we don’t mind the lazier answer, “half the supply voltage.” This will be our usual answer, for single-supply circuits.
- ¹⁴ To calculate this you’ll need to decide what is the effective *resistance* that should be paired with the blocking capacitor. C and that R form a highpass, as you know.
- ¹⁵ Yes, just R_C . You’ve seen this point explained in Chapter 4N.

4L.5.1 Maximizing gain: sneak preview of Ebers–Moll at work

Now let's try a case that our first, simple, view of the common-emitter amplifier does not describe accurately: check out Fig. 4L.7 to see what happens if we parallel the emitter resistor with a big capacitor, so that at “signal frequencies” R_E is shorted out.

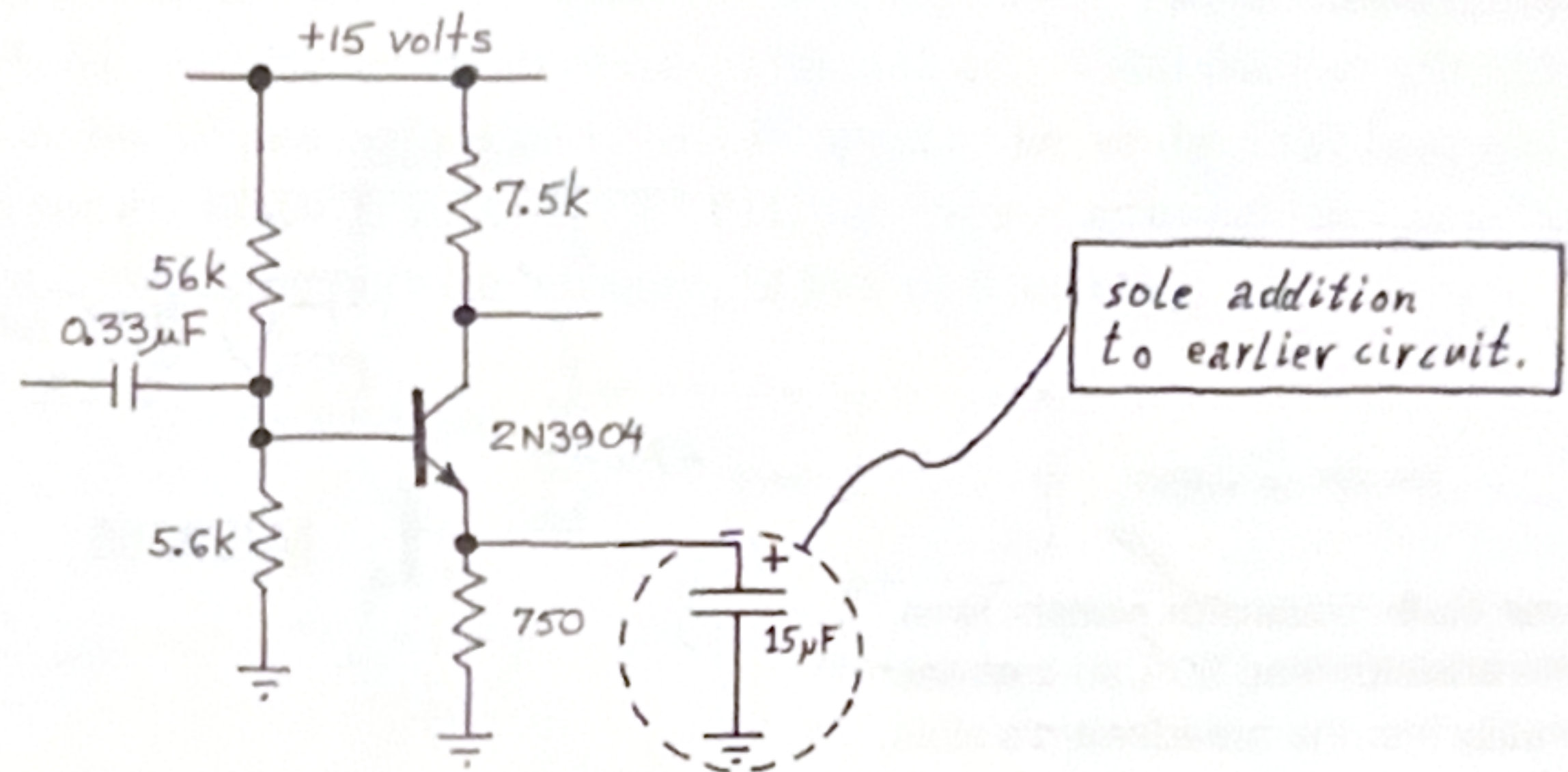


Figure 4L.7 Grounded emitter amplifier.

Modify your common-emitter amp to make the amplifier shown in Fig. 4L.7 (similar to Fig. 5N.17). What circuit properties does your addition of the *bypassing* capacitor affect? It affects gain, because R_E disappears from the gain equation. Does the C alter the biasing of the output (which was designed to be centered roughly at half the supply voltage)?¹⁶

Now drive this new circuit with a small *triangle* wave at 10 kHz, at an amplitude that almost produces clipping (you'll need to use plenty of attenuation – 40 dB or more – in the function generator). Does the output waveform look like Fig. 4L.8? (Compare to Fig. 5N.12.)

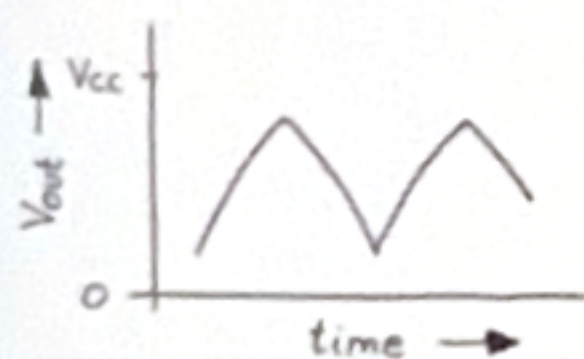


Figure 4L.8 Large-swing output of grounded emitter amplifier when driven by a triangle wave.

Installing the cap that bypasses R_E does *not* deliver infinite gain: sorry! In Chapter 5N you'll discover that we can predict the gain by adding to our transistor model a little resistor-like element in the emitter, a modelling device that we call (affectionately) “little r_e .” In the present circuit, where we're running a quiescent current of about 1 mA, little r_e takes a value of about 25 Ω . See if your circuit's gain is consistent with that value for r_e .

This measurement will be difficult. First, you will need to reduce the function generator output to a level close to the minimum possible. Then you will find the input is immeasurably small, viewed with a 10 $\times$ probe. This, then, is one of the rare occasions when you need to revert to a 1 $\times$ display: use a BNC rather than a 10 $\times$ probe as you watch the *input* signal.¹⁷

Typically, the gain you observe is around 250 – lower than you predicted. Low I_C can help explain that; so can any curvature that remains in the output waveform: the gain revealed by such a waveform is sometimes higher than the quiescent value, sometimes lower, but the *net* effect of the curvature is to *reduce* gain.

4L.6 Transistor switch

The circuit in Fig. 4L.9 differs from all those you have built so far: the transistor, when ON, is *saturated*. In this regime you should not expect to see $I_C = \beta I_B$. Why not?¹⁸ In addition, β for this big power

¹⁶ No: biasing is a simple DC effect.

¹⁷ If your scope probe has a 10 $\times$ /1 $\times$ selector switch, just select 1 $\times$.

¹⁸ One way to explain this is just to note that achieving a current as high as that would require either a smaller resistance on the collector or a larger power supply. When the transistor is saturated, the current is limited not by the transistor but by the *load*.

transistor is much lower than what you have seen for the *small signal* transistor, 2N3904 – at least at high currents. Minimum β for the MJE3055T is a mere 20, at $I_C=4A$. But the transistor is big, and lives in a big package (“TO-220”), a package that can dissipate a lot of heat, especially if the metal package is thermally attached to a still bigger piece of metal (a “heat sink”). The large size of the transistor itself keeps current density down and saturation voltage low. You’ll measure that $V_{CE(sat)}$ in a few minutes.

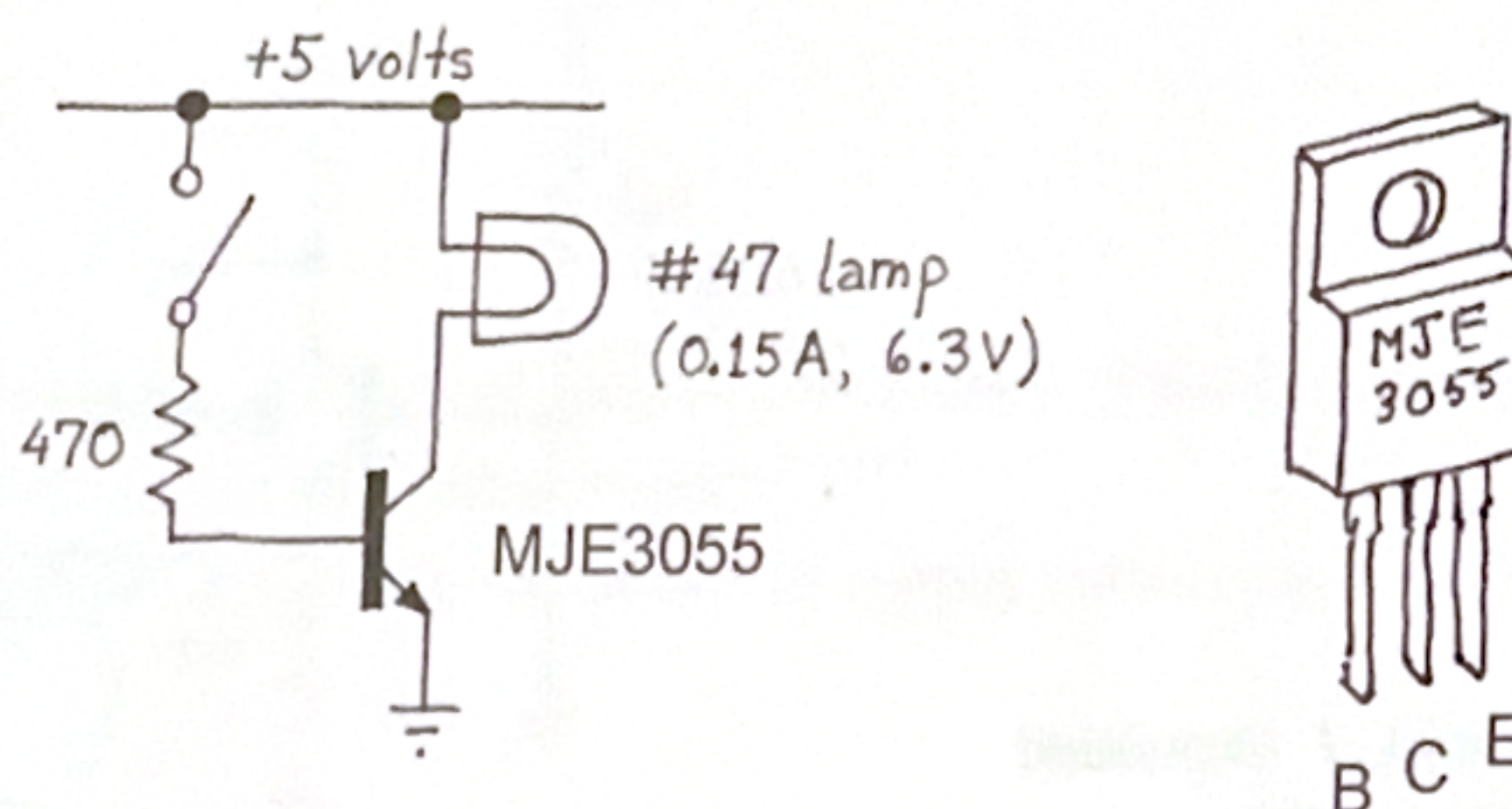


Figure 4L.9 Transistor switch: twist each transistor lead 90°, so that each fits easily into the breadboard's slots.

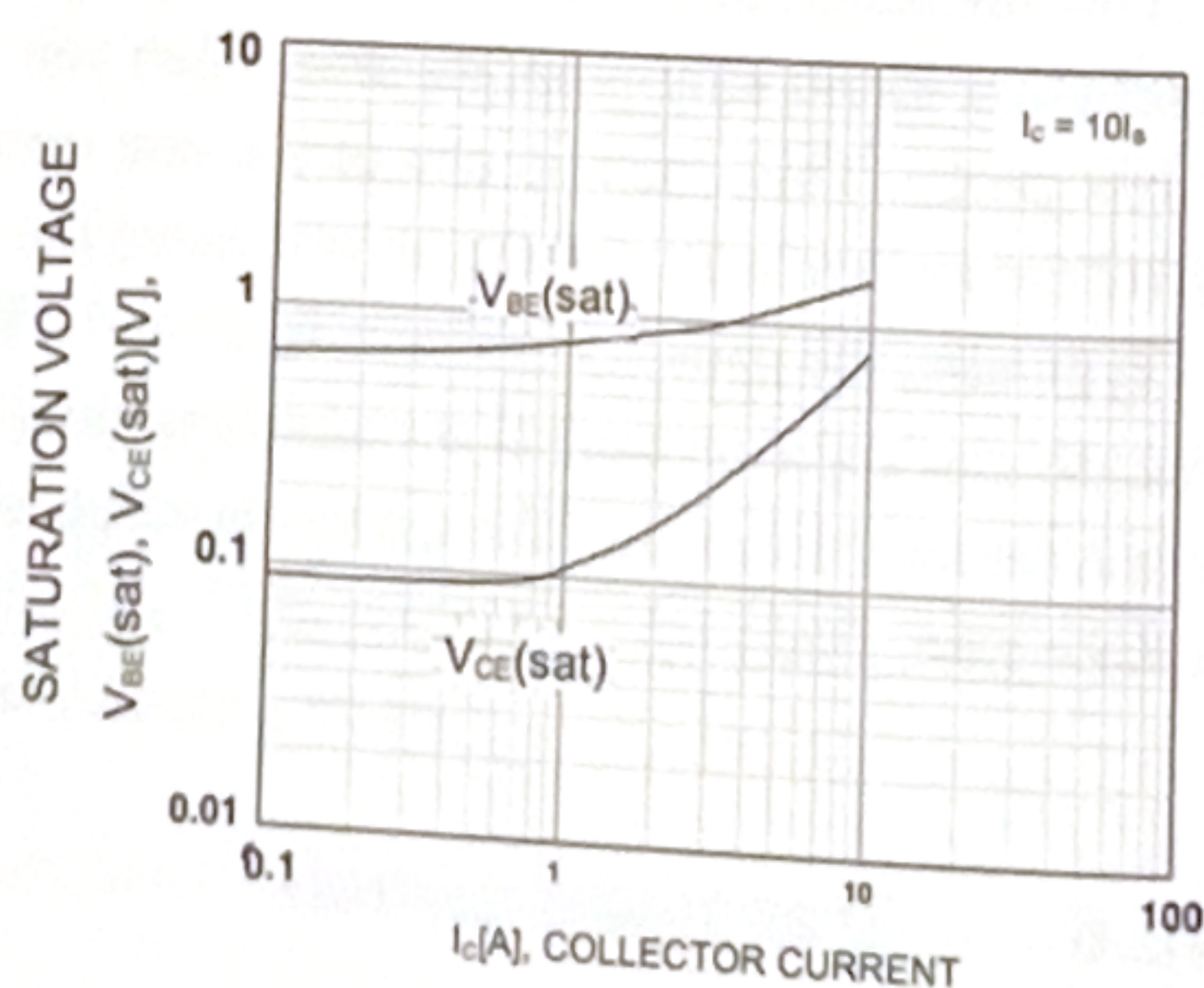
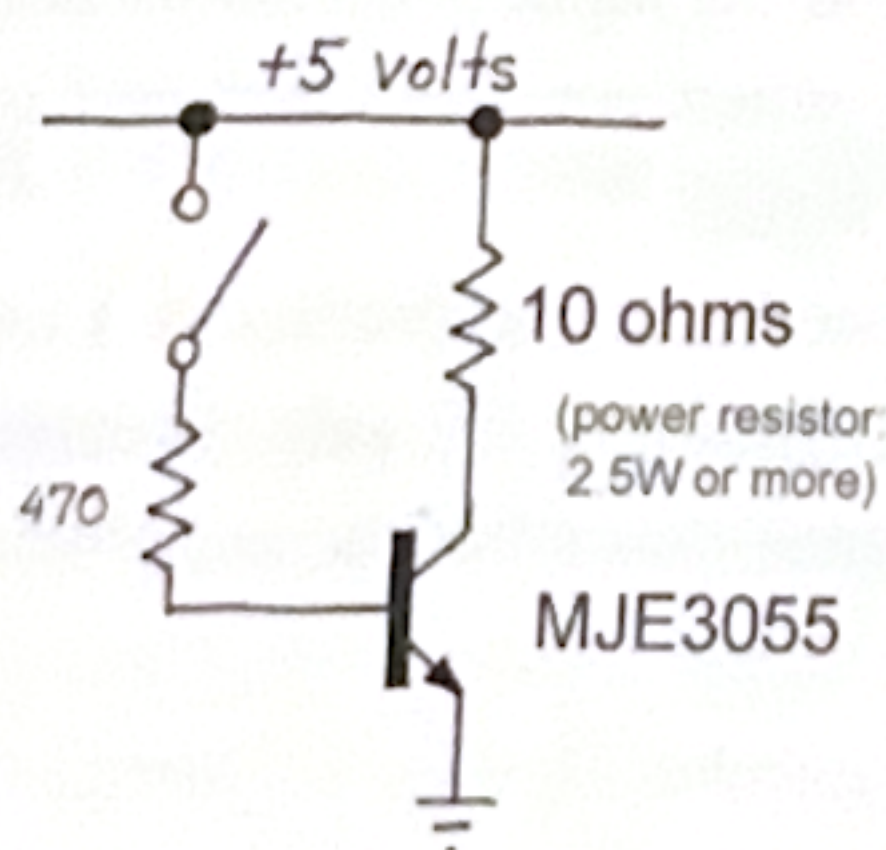
Turn the base current on and off by flipping the toggle switch – or, if you’re too lazy to wire a switch, by pulling one end of the resistor out of the breadboard and touching it either to +5 or to ground. What is I_B , roughly? What is the minimum required β ?¹⁹

4L.6.1 Saturation or “On” voltage: $V_{CE(sat)}$

Now let’s make the transistor work harder. Replace the lamp with a 10 Ω *power* resistor (not an ordinary 1/4 W resistor; consider how much power the resistor must dissipate: $V^2/R = 25/10 = 2.5$ W). A 1/4 W resistor would cook in this circuit.

Measure the saturation voltage, $V_{CE(sat)}$, with DVM or scope. Then parallel the base resistor with 150 Ω , and note the improved $V_{CE(sat)}$. Compare your results with the results promised by the data set out in Fig. 4L.10. Note that the curves assume heavy base drive: $I_B = I_C/10$, not I_C/β .

Figure 4L.10 Switch saturation: a heavier load current asks more of the switch here (data courtesy of ON Semiconductor).



We will return to transistor switches later, in Lab 12L, when we’ll meet the leading competitor for the tasks a ’3055 can do: a big power field-effect transistor (“MOSFET”). At that time, we’ll set up a

¹⁹ You’ll notice that we are *overdriving* the base, here, as is usual in a switch. Since base current here is about 10 mA, even at the specified minimum *beta* of 20, the transistor could pass more current than the load will permit. That’s good: the switch will be well saturated, and its V_{CE} will therefore be low.

competition between a '3055 and a MOSFET and see which does a better job of delivering power to the load.

4L.6.2 Switch an inductive load (more exciting)

Now replace the resistor at the collector with a 10mH inductor as in Fig. 4L.11. Replace the 5 V supply with 1.5 V from a single AA cell (we don't want to overheat the inductor – and the low cell voltage makes all the more impressive the voltage spike that soon will appear). Drive the 1 k Ω base resistor with a 1 kHz square wave from the function generator's "TTL" or "Sync" terminal (this is a "logic level" waveform: a square wave switching between ground and four or five volts).

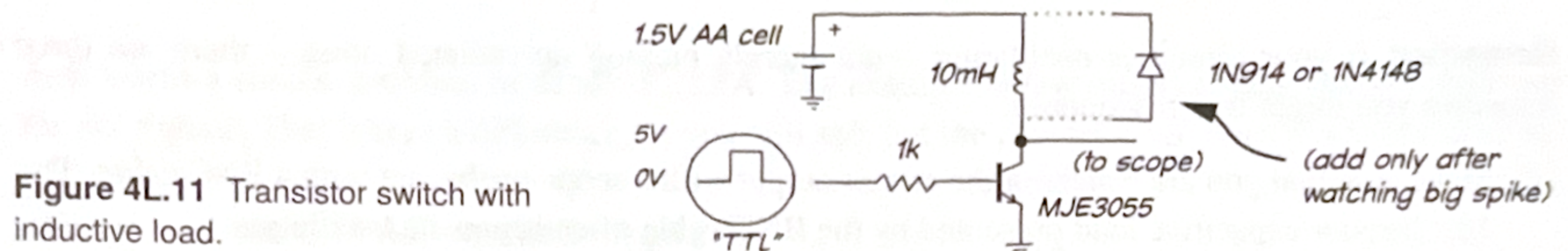


Figure 4L.11 Transistor switch with inductive load.

At the transistor's collector, you should find the inductor giving the transistor an alarming voltage spike (we saw more than 100V). When the transistor tries to turn off, the inductor tries to keep the current flowing. Its method is to drive the collector voltage up, until the transistor "breaks down," permitting the inductor to have its way – keeping the current flowing despite the transistor's attempt to turn it off abruptly.

A diode from collector to V_+ tames this voltage spike; such a diode clamp is a standard protection in circuits that switch inductive loads. Most transistors don't like to break down (this time we didn't mind since our goal was to show you this spike). Incidentally, we'll see a circuit that makes good use of this voltage spike when we look at switching power supplies later in Lab 12L.

4L.7 A note on power-supply noise

Note §9S.3 is devoted to noise problems. Here, we offer just a few pointers related to one form of noise.

In Figure 4L.12 is one of today's followers, swept rather slowly (at 20 μ s/division). The follower has been fed a sine of about 15 kHz from the function generator. A strange *thickening* of the trace appears.

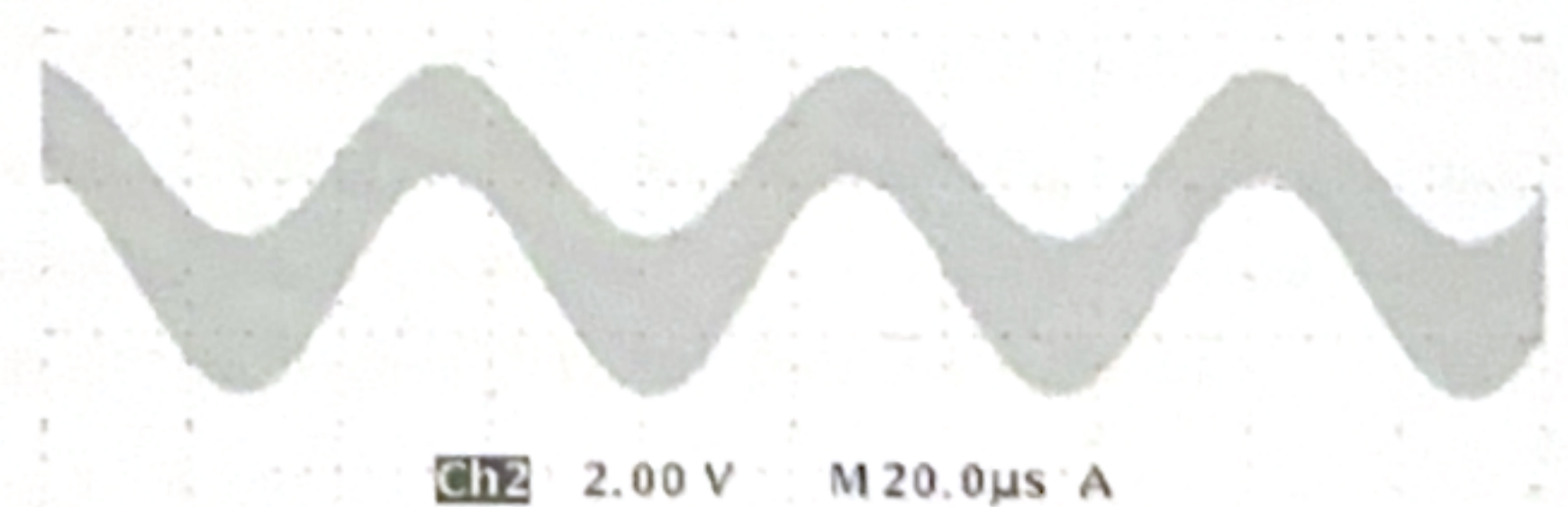


Figure 4L.12 Thickening of scope trace indicates a nasty "parasitic" oscillation.

Sweeping the scope faster (Fig. 4L.13, at 10 ns/div) resolves the "thickening" into a very fast sinusoid – up in the *FM* radio broadcast range. Decoupling the power supplies should eliminate this problem (see below).

Oscillations versus radio pickup

If you see this fuzz, try turning off your breadboard power supply. If the fuzz disappears, your circuit is guilty: it was causing the fuzz, by running as an unintended *oscillator* (we'll look closely at the